

The `mrDi` File Format

Antony Della Vecchia, Michael Joswig and Benjamin Lorenz

TU Berlin

2026-07-01



Mathematical Research Data Initiative — part of Germany's NFDI, building research-data infrastructure for mathematics.



Mathematical Research Data Initiative — part of Germany's NFDI, building research-data infrastructure for mathematics.

The **Computer Algebra** task area, led by PIs **Michael Joswig** (TU Berlin) and **Claus Fieker** (RPTU Kaiserslautern), runs four projects:



Mathematical Research Data Initiative — part of Germany's NFDI, building research-data infrastructure for mathematics.

The **Computer Algebra** task area, led by PIs **Michael Joswig** (TU Berlin) and **Claus Fieker** (RPTU Kaiserslautern), runs four projects:

- **File format** — Antony Della Vecchia (TU Berlin)
- **Guidelines for reproducible workflows** — Lars Kastner (TU Berlin)
- **Technical peer review** — Jeroen Hanselmann (RPTU Kaiserslautern)
- **Containerization** — Aaruni Kaushik (RPTU Kaiserslautern)



Infrastructure

- In mathematics, a theorem from another field is never off-limits: if it applies to your work, you use it.



Infrastructure

- In mathematics, a theorem from another field is never off-limits: if it applies to your work, you use it.
- Software should be the same — a result computed in one system should not be locked out of another.



Infrastructure

- In mathematics, a theorem from another field is never off-limits: if it applies to your work, you use it.
- Software should be the same — a result computed in one system should not be locked out of another.
- To support these ideas we need infrastructure.
- Software infrastructure is important for research — but is often overlooked and underfunded.



Infrastructure

- In mathematics, a theorem from another field is never off-limits: if it applies to your work, you use it.
- Software should be the same — a result computed in one system should not be locked out of another.
- To support these ideas we need infrastructure.
- Software infrastructure is important for research — but is often overlooked and underfunded.
- The tools mathematicians compute with deserve the same care as the mathematics itself.



Infrastructure

- In mathematics, a theorem from another field is never off-limits: if it applies to your work, you use it.
- Software should be the same — a result computed in one system should not be locked out of another.
- To support these ideas we need infrastructure.
- Software infrastructure is important for research — but is often overlooked and underfunded.
- The tools mathematicians compute with deserve the same care as the mathematics itself.



Why Files?

- People often have a preferred software system.



Why Files?

- People often have a preferred software system.
- Computations can be expensive.



Why Files?

- People often have a preferred software system.
- Computations can be expensive.
- Verification of results is at most as computationally expensive.



Why Files?

- People often have a preferred software system.
- Computations can be expensive.
- Verification of results is at most as computationally expensive.
- Software changes often.
- Example: Surfaces in $\mathbb{P}^4$



A History of File Formats

```
* Class:      LP
* Rows:       5
* Columns:    2
* Format:     MPS
*
Name          unnamed#0
ROWS
N  C0000000
...
G  R0000003
COLUMNS
  x1  C0000000  2      R0000000  1
  x1  R0000001 -1
  x2  C0000000  3      R0000002  1
  x2  R0000003 -1
RHS
  B  C0000000 -1      R0000000 -1
  B  R0000001 -1      R0000002 -1
  B  R0000003 -1
BOUNDS
FR BND      x1
FR BND      x2
ENDATA
```

- The LP and MPS file formats. IBM [1970s] (industry standards)
- Mathematica Notebooks. Wolfram [1988]
- OpenMath (tree structure). Mike Dewar [2000]
- IPython 0.12 interactive browser notebooks (Jupyter) [2011]
- polymake file format. Gawrilow, Hampe, Joswig [2016]



Storing Computer Algebra Data

- It's common to have multiple perspectives on an object in mathematics.
- While storing mathematical data a choice of perspective must be made.
- Such a choice might not be describable in an email.

Say we want to store:

$$p = 2y^3z^4 + (a + 3)z^2 + 5ay + 1$$



Storing Computer Algebra Data

- It's common to have multiple perspectives on an object in mathematics.
- While storing mathematical data a choice of perspective must be made.
- Such a choice might not be describable in an email.

Say we want to store:

$$p = 2y^3z^4 + (a + 3)z^2 + 5ay + 1$$

- Some technicalities with the coefficients.
- Is y considered a coefficient of z ?
- What is a ?



Storing Computer Algebra Data

- It's common to have multiple perspectives on an object in mathematics.
- While storing mathematical data a choice of perspective must be made.
- Such a choice might not be describable in an email.

Say we want to store:

$$p = 2y^3z^4 + (a + 3)z^2 + 5ay + 1$$

- Some technicalities with the coefficients.
- Is y considered a coefficient of z ?
- What is a ?
- How can we guarantee the objects behave as expected on load?



OpenMath: A Top-Down Approach

- Goal: a software-independent standard for the meaning of mathematical objects, not just their syntax.



OpenMath: A Top-Down Approach

- Goal: a software-independent standard for the meaning of mathematical objects, not just their syntax.
- An OpenMath object is a tree of symbols, each defined in a **Content Dictionary**.



OpenMath: A Top-Down Approach

- Goal: a software-independent standard for the meaning of mathematical objects, not just their syntax.
- An OpenMath object is a tree of symbols, each defined in a **Content Dictionary**.
- Top-down: agree on the semantics of all of mathematics first, then every system speaks the same language.



OpenMath: A Top-Down Approach

- Goal: a software-independent standard for the meaning of mathematical objects, not just their syntax.
- An OpenMath object is a tree of symbols, each defined in a **Content Dictionary**.
- Top-down: agree on the semantics of all of mathematics first, then every system speaks the same language.

Fateman's critique:

- Formalising the semantics of all mathematics is open-ended, and meaning is context-dependent — there is no single agreed semantics.



OpenMath: A Top-Down Approach

- Goal: a software-independent standard for the meaning of mathematical objects, not just their syntax.
- An OpenMath object is a tree of symbols, each defined in a **Content Dictionary**.
- Top-down: agree on the semantics of all of mathematics first, then every system speaks the same language.

Fateman's critique:

- Formalising the semantics of all mathematics is open-ended, and meaning is context-dependent — there is no single agreed semantics.
- Using the semantics means reimplementing every Content Dictionary — in effect, building a whole separate computer algebra system to read a file.

R. Fateman, [A Critique of OpenMath and Thoughts on Encoding Mathematics](#).



A Bottom Up Approach

What is a bottom-up approach?



A Bottom Up Approach

What is a bottom-up approach?

- No attempt to formalise the language of mathematics up front.



A Bottom Up Approach

What is a bottom-up approach?

- No attempt to formalise the language of mathematics up front.
- Semantics come from the software that produces the data, not from a global specification.



A Bottom Up Approach

What is a bottom-up approach?

- No attempt to formalise the language of mathematics up front.
- Semantics come from the software that produces the data, not from a global specification.
- The format grows from real data and real software, rather than the other way around.



A Bottom Up Approach

What is a bottom-up approach?

- No attempt to formalise the language of mathematics up front.
- Semantics come from the software that produces the data, not from a global specification.
- The format grows from real data and real software, rather than the other way around.

What features do we need to support it?



A Bottom Up Approach

What is a bottom-up approach?

- No attempt to formalise the language of mathematics up front.
- Semantics come from the software that produces the data, not from a global specification.
- The format grows from real data and real software, rather than the other way around.

What features do we need to support it?

- **Namespaces** to pin semantics to a specific software and version.



A Bottom Up Approach

What is a bottom-up approach?

- No attempt to formalise the language of mathematics up front.
- Semantics come from the software that produces the data, not from a global specification.
- The format grows from real data and real software, rather than the other way around.

What features do we need to support it?

- **Namespaces** to pin semantics to a specific software and version.
- **Open source implementations** of saving and loading.



A Bottom Up Approach

What is a bottom-up approach?

- No attempt to formalise the language of mathematics up front.
- Semantics come from the software that produces the data, not from a global specification.
- The format grows from real data and real software, rather than the other way around.

What features do we need to support it?

- **Namespaces** to pin semantics to a specific software and version.
- **Open source implementations** of saving and loading.
- **Versioning and upgrades**: older files are upgraded on load.



A Bottom Up Approach

What is a bottom-up approach?

- No attempt to formalise the language of mathematics up front.
- Semantics come from the software that produces the data, not from a global specification.
- The format grows from real data and real software, rather than the other way around.

What features do we need to support it?

- **Namespaces** to pin semantics to a specific software and version.
- **Open source implementations** of saving and loading.
- **Versioning and upgrades**: older files are upgraded on load.



The mrdi File Format

- Based on JSON (tree structure).



The `mrdi` File Format

- Based on JSON (tree structure).
- Generalizes ideas from the `polymake` format to algebraic data.



The `mrdr` File Format

- Based on JSON (tree structure).
- Generalizes ideas from the `polymake` format to algebraic data.
- Namespaces pin semantics to a specific software.



The `mrDI` File Format

- Based on JSON (tree structure).
- Generalizes ideas from the `polymake` format to algebraic data.
- Namespaces pin semantics to a specific software.
- References to shared parent types, stored with UUIDs.

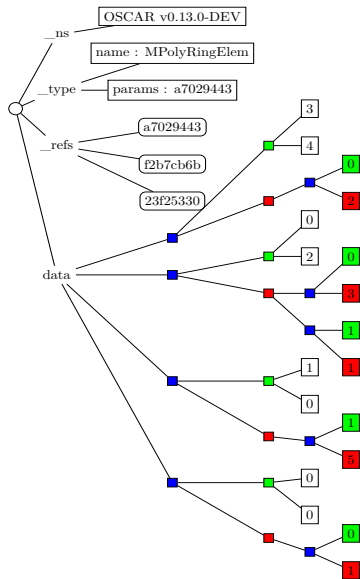


The `mrDI` File Format

- Based on JSON (tree structure).
- Generalizes ideas from the `polymake` format to algebraic data.
- Namespaces pin semantics to a specific software.
- References to shared parent types, stored with UUIDs.
- First version developed using OSCAR.



Tree Structure



$$2y^3z^4$$

$$(a + 3)z^2$$

$$5ay$$

$$1$$



Example File Serialized with OSCAR

```
{
  "_ns": { "Oscar": [ "https://github.com/oscar-system/Oscar.jl", "1.0.0" ] },
  "_type": {
    "name": "MPolyRingElem",
    "params": "869a359a-43d3-43f4-9821-0af9346be019"
  },
  "data": [[["3", "4"], [{"0", "2"}]],
    [{"0", "2"}, [{"0", "3"}, ["1", "1"]]],
    [{"1", "0"}, [{"1", "5"}]],
    [{"0", "0"}, [{"0", "1"}]]],
  "_refs": {
    "152ac7bd-e85a-4b36-acc2-743ade2cad4f": {
      "data": { "base_ring": { "data": "7", "_type": "FqField"},
        "symbols": ["x"] },
      "_type": "PolyRing"
    },
    "869a359a-43d3-43f4-9821-0af9346be019": {
      "data": {
        "base_ring": "a8309b96-caec-443c-bedb-e23bb0634c14",
        "symbols": [ "y", "z" ]
      },
      "_type": "MPolyRing" },
    "a8309b96-caec-443c-bedb-e23bb0634c14": {
      "data": {
        "def_pol": {
          "data": [{"0", "1"}, ["2", "1"]],
          "_type": {
            "name": "PolyRingElem",
            "params": "152ac7bd-e85a-4b36-acc2-743ade2cad4f"
          }
        }
      },
      "_type": "FqField"
    }
  }
}
```



Shared Parents Across Sessions

Save two polynomials — the shared ring is stored once, referenced by UUID:

```
julia> R, (y, z) = polynomial_ring(QQ, [:y, :z]);
```

```
julia> save("p.mrdi", 2*y^3*z^4 + 3*z^2 + 5*y + 1)
```

```
julia> save("q.mrdi", y + 1)
```



Shared Parents Across Sessions

Save two polynomials — the shared ring is stored once, referenced by UUID:

```
julia> R, (y, z) = polynomial_ring(QQ, [:y, :z]);  
julia> save("p.mrdi", 2*y^3*z^4 + 3*z^2 + 5*y + 1)  
julia> save("q.mrdi", y + 1)
```

In a **new session**, load both and multiply — they share a parent:

```
julia> p = load("p.mrdi");  
julia> q = load("q.mrdi");  
  
julia> parent(p) === parent(q)  
true  
  
julia> p * q  
2*y^4*z^5 + 2*y^3*z^4 + 3*y*z^3 + 5*y^2*z + 3*z^2 + y*z + 5*y + 1
```



Saving and Loading in OSCAR

Save an object to a mrdi file and load it back:

```
julia> using Oscar  
  
julia> R, (y, z) = polynomial_ring(QQ, [:y, :z]);  
  
julia> save("poly.mrdi", 2*y^3*z^4 + 3*z^2 + 5*y + 1)  
  
julia> load("poly.mrdi")  
2*y^3*z^4 + 3*z^2 + 5*y + 1
```



Saving and Loading in OSCAR

Save an object to a `mr di` file and load it back:

```
julia> using Oscar  
  
julia> R, (y, z) = polynomial_ring(QQ, [:y, :z]);  
  
julia> save("poly.mr di", 2*y^3*z^4 + 3*z^2 + 5*y + 1)  
  
julia> load("poly.mr di")  
2*y^3*z^4 + 3*z^2 + 5*y + 1
```

Override the parent on load — reinterpret the same data in a new ring:

```
julia> S, (u, v) = polynomial_ring(GF(7), [:u, :v]);  
  
julia> load("poly.mr di"; params=S)  
2*u^3*v^4 + 3*v^2 + 5*u + 1
```



Where We Are Today

- All core OSCAR types can be serialized.



Where We Are Today

- All core OSCAR types can be serialized.
- Multiple upgrades to the format.



Where We Are Today

- All core OSCAR types can be serialized.
- Multiple upgrades to the format.
- Multiprocess computing backed by the format.



Where We Are Today

- All core OSCAR types can be serialized.
- Multiple upgrades to the format.
- Multiprocess computing backed by the format.
- Growing interest from other computer algebra systems:



Where We Are Today

- All core OSCAR types can be serialized.
- Multiple upgrades to the format.
- Multiprocess computing backed by the format.
- Growing interest from other computer algebra systems:
 - Macaulay2 1.26 ships with a MRDI package (Doug Torrance):
saveMRDI / loadMRDI for $\mathbb{Z}$, $\mathbb{Q}$, finite and Galois fields, polynomial rings, ideals, matrices.



Where We Are Today

- All core OSCAR types can be serialized.
- Multiple upgrades to the format.
- Multiprocess computing backed by the format.
- Growing interest from other computer algebra systems:
 - Macaulay2 1.26 ships with a MRDI package (Doug Torrance):
saveMRDI / loadMRDI for $\mathbb{Z}$, $\mathbb{Q}$, finite and Galois fields, polynomial rings, ideals, matrices.
 - A Magma implementation by Jeroen Hanselman (MagmaMaRDI-JSON):
booleans, strings, $\mathbb{Z}$, $\mathbb{Q}$, finite fields, polynomial rings, matrices, arrays.



Where We Are Today

- All core OSCAR types can be serialized.
- Multiple upgrades to the format.
- Multiprocess computing backed by the format.
- Growing interest from other computer algebra systems:
 - Macaulay2 1.26 ships with a MRDI package (Doug Torrance):
saveMRDI / loadMRDI for $\mathbb{Z}$, $\mathbb{Q}$, finite and Galois fields, polynomial rings, ideals, matrices.
 - A Magma implementation by Jeroen Hanselman (MagmaMaRDI-JSON):
booleans, strings, $\mathbb{Z}$, $\mathbb{Q}$, finite fields, polynomial rings, matrices, arrays.
 - A SageMath implementation (SageMathMRDIJSON, built with Claude Code): matrices, polynomial rings and ideals over $\mathbb{Q}$, compatible with OSCAR's format.
 - Some Lean functionality (Cedric Holler, Doug Torrance)



Databases and Collections

OscarDB — collections in a MongoDB, queried directly from OSCAR; a RESTful API is coming soon.



Databases and Collections

OscarDB — collections in a MongoDB, queried directly from OSCAR; a RESTful API is coming soon.

- SmallTreeModels
- SelfProjectingMatroids
- LeechPairs
- TransitiveSimplicialComplexes



Databases and Collections

OscarDB — collections in a MongoDB, queried directly from OSCAR; a RESTful API is coming soon.

- SmallTreeModels
- SelfProjectingMatroids
- LeechPairs
- TransitiveSimplicialComplexes

Collections shipped with OSCAR

- ArchimedeanSolids, CatalanSolids, JohnsonSolids
- Surfaces in $\mathbb{P}^4$, TauTaubarGenericEnriquesSurfaces



Thank you!

